Outline ESYS 104: The Earth System 2026

Earth can be viewed as a dynamic interacting system of atmosphere, life, ocean, rock and soil. Earth is open energetically but largely closed materially. Earth's surface and interior have changed through time and continue to change today. Understanding these changes and what drives them is the goal of Earth system science. This course will introduce the core processes of the 'Earth System' and apply them to the 21st Century global challenges posed by climate change, which is occurring at the same time as increasing stresses on services provided by natural ecosystems due to human population growth and resource utilization.

This course is aimed at the U0 Science level; hence we will build upon your existing math and science knowledge.

Time: Monday-Wednesday, 11:25 AM – 12:55 PM in person ***Stewart Biology S1/3***

Instructor:

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Teaching Assistants (email on *myCourses*)***: Yan Su, Jordan Thakar, Bonnie De Baets, William Fajzel***

Course Topics:

No.	*Date*	*Subject*	
1.	31-Aug	Introduction	
2.	02-Sep	Solar system/Earth materials	
	07-Sep	**Labour Day**	
3.	09-Sep	Plate tectonics	
4.	14-Sep	Reservoirs and resources	
5.	16-Sep	Energy and the atmosphere	Quiz #1
6.	21-Sep	Atmospheric circulation	
7.	23-Sep	Ocean circulation/Arctic Ocean	
8.	28-Sep	Hydrology, soils and their productivity	Assig #1 due
9.	30-Sep	Biomes, ecology and ecosystems	Quiz #2
	05-Oct	**Classes cancelled** -Quebec elections	
10.	07-Oct	Ecosystem resilience and regime shifts	Feedbacks question
	12-Oct	**Fall Break**	
	14-Oct	**Fall Break**	
11.	19-Oct	Deep time/Earth history	Assig #2 due
12.	21 Oct	Himalayas, weathering	
13.	26-Oct	**MIDTERM exam** (in class)	
14.	28-Oct	Snowball Earth (evolution of atmosphere)	

No.	*Date*	*Subject*	
15.	02-Nov	Glacial Cycles	Quiz #3
16.	04-Nov	Global Climate Change & PaleoClimate	Proxy question
17.	09-Nov	Arctic/Antarctic Climate Change	
18.	11-Nov	Global population growth	
19.	16-Nov	The Sahel and the subtropics	Assig #3 due
20.	18-Nov	Coastal Processes	
21.	23-Nov	Earth System hazards	Quiz #4
22.	25-Nov	Fossil Fuels	
23.	30-Nov	Energy alternatives	Assig #4 due
24.	02-Dec	Planetary Boundaries	Quiz #5
25.	03-Dec	Tipping Points in the Climate System; Review	

Course topics and order may still be adjusted; assessments are fixed. There is no textbook, but required readings will be announced in class and posted on *myCourses.*

Marking: 40% assignments, 15% quizzes, short writings, in-class exercises, 20% mid-term, 25% Final Exam. The supplemental final exam (given in May) will count for 25% of the grade.

There will be four assignments throughout the course and a comprehensive final exam (similar in format to the assignments/quizzes). You may work together on assignments, but what you turn in must be your own work in your own words. **Late assignments** are given a penalty of 10% per day. There will be ample time to complete the assignments; it's important that feedback can be given promptly. One of the in class exercises can be dropped without penalty.

Lectures will be recorded. There will be a number of samples to observe, demonstrations, and in-class exercises, so attendance is necessary for the full experience.

myCourses
myCourses will be used extensively through the course. *myCourses* has the e-mail addresses for the teaching assistants, as well as a Group Discussion section; this is the second best place (the first is in class) to ask questions on the course material. We will also be posting additional readings, links to Earth Systems web sites, lecture material, and study guides on *myCourses*. Check *myCourses* frequently for course announcements. There will also be a social media site dedicated to the course.

Additional Information:
"McGill University values **academic integrity**. Therefore, all students must understand the meaning and consequences of cheating, plagiarism and other academic offences under the Code of Student Conduct and Disciplinary Procedures (see www.mcgill.ca/students/srr/honest/ for more information)." Work submitted for evaluation as part of this course may be checked with text matching software within MyCourses.

"In accord with McGill University's Charter of Students' Rights, students in this course have the right to submit in **English** or in **French** any written work that is to be graded."

In the event of extraordinary circumstances beyond the University's control, the evaluation scheme in a Course is subject to change, provided that there be timely communications to the students regarding the change.

Use of Large Language Models (LLMs) (Copilot, Claude, Gemini, ChatGPT, Grok, etc.):

LLMs show up at the top of many searches, and McGill has contracted with Microsoft to use Copilot. Hence they are very hard to ignore.

Experience has shown that students that use LLMs to complete their assignments end up doing poorly on the in-class exams. You've been warned.

Recording in the classroom (except for the lecture recordings) is not permitted.

This includes any use of Smart Glasses (Meta, etc.). It is relatively easy to disable the recording light on these glasses, so if you must ware a pair of smart glasses to class, you **MUST** put a piece of opaque tape over the camera and microphone.

Appendix 1. General Science Grading Rubric

Score	**Identification of relevant concepts** (Choice of correct model, theory, equation, etc.);	**Correct application of concepts** (Correct combination and application of models, theories, equations, etc.)	**Efficiency of approach** (No extra steps or extraneous information given)	**Quality of presentation** (Clarity, language, nomenclature, citing specific sources or examples where appropriate)
0	Concepts identified are completely irrelevant, or no concepts identified at all.	Concepts are applied completely incorrectly, or no attempt has been made to apply concepts.	The entirety of the work presented is unnecessary or irrelevant, or no approach has been taken at all.	Work is unclear and fails to use appropriate nomenclature. Citations (where required) are absent.
1	Some identified concepts are at least partly correct. Important concepts are missing and/or incorrect concepts are identified	Some concepts have been combined or applied in a partially appropriate manner. Important steps or syntheses are missing and/or incorrect steps are taken.	Much of the work presented is unnecessary or irrelevant.	Work is largely unclear and only occasionally uses appropriate nomenclature. Citations (where required) are substandard.
2	The identified concepts are largely correct and partly complete.	The application of concepts is somewhat appropriate with multiple minor, or a few major errors.	Some unnecessary steps are taken and/or unnecessary information is given.	Work is partly clear and uses some appropriate nomenclature. Citations (where required) are substandard.
3	The identified concepts are largely correct and mostly complete.	The application of concepts is largely appropriate with no major errors and few minor ones.	Few unnecessary steps are taken and/or unnecessary information is given.	Work is generally clear and uses appropriate nomenclature. Citations (where required) are appropriate.
4	All of the relevant concepts are identified, and no incorrect concepts are chosen.	The application of concepts is entirely correct and error-free.	No unnecessary steps are taken and no unnecessary information is given.	Work is at the highest level of clarity, using entirely appropriate nomenclature. Citations (where required) are comprehensive.